

Implementing random fields

The goal of this exercise session is to implement random fields by means of the Karhunen-Loève expansion, and apply it to stochastic finite element analysis.

1 Implementing random fields by the Karhunen-Loève expansion

Consider a one-dimensional Gaussian random field with zero mean and covariance function $Cov(x_1, x_2) = \exp\left(-\frac{|x_1 - x_2|}{l_x}\right)$, $x_1, x_2 \in [0, 1]$, let us approximate the random field by the Karhunen-Loève expansion:

1. Solve the Karhunen-Loève expansion using analytical methods.
2. Solve the Karhunen-Loève expansion using numerical methods, e.g., the Galerkin method.

2 Investigate stochastic finite element analysis

3. Assemble the stochastic finite element equation based on the obtained random field, e.g., for a one-dimensional elastic beam.
4. Solve the obtained stochastic finite element equation using the Monte Carlo simulation.

3 Investigate two-dimensional cases

5. Extend the above implementations to two-dimensional problems, including two-dimensional random fields and the corresponding stochastic finite element analysis, e.g., a two-dimensional elastic beam with random Young's modulus.